# TASK 1 – BASIC NETWORK SCANNING WITH NMAP

**Track:** Cybersecurity

**Level / Number:** Level 1 – Task 1

**Task Title:** Basic Network Scanning with Nmap

## Objective

Perform a basic network scan using Nmap to identify open ports and services on an authorized local computer or virtual machine. Identify service versions and operating-system information and analyze the security risks associated with open ports.

## Tech Stack / Tools

Nmap, Kali Linux, Linux Terminal, VirtualBox, Local Virtual Machine

## Feature Checklist

**1. Install Nmap**

nmap --version
If needed: sudo apt update

sudo apt install nmap

**2. Find Target IP**

ip addr
Find the IPv4 address of your authorized local machine or VM.

**3. Basic Scan**

nmap YOUR-IP

## Basic Scan Screenshot

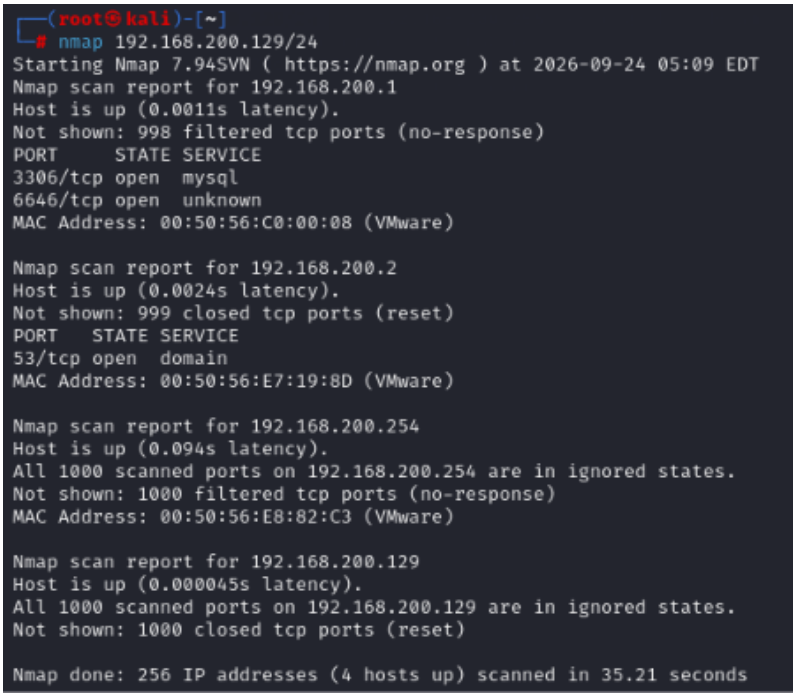



Figure 1. Basic Nmap scan of the 192.168.200.0/24 local network.

## 4. Service Version Scan

Command: **nmap -sV YOUR-IP**

The service/version scan identifies the services associated with open ports and attempts to determine their versions.

## Service / Version Scan Screenshot

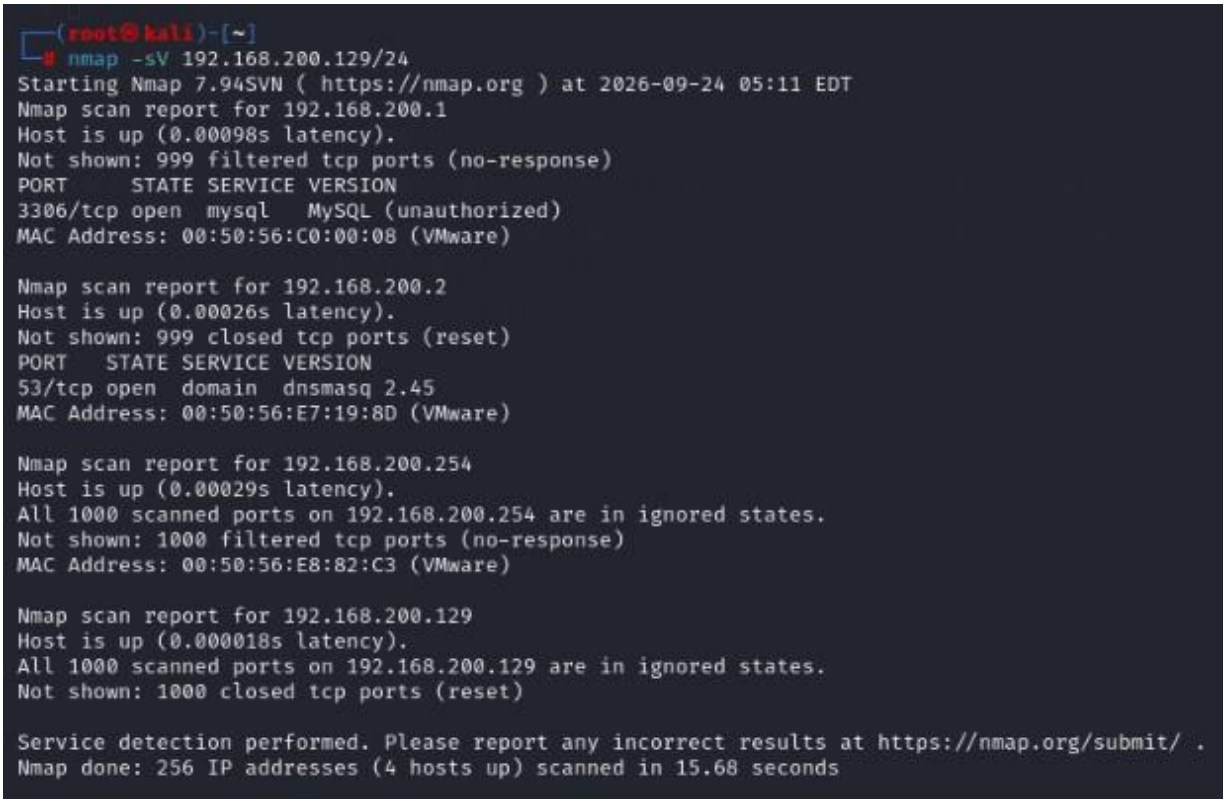



Figure 2. Service/version detection results. TCP 3306 is identified as MySQL on 192.168.200.129, and TCP 53 is identified as domain/dnsmasq 2.45 on 192.168.200.2.

## 5. OS Detection

Command: **sudo nmap -O YOUR-IP**

The OS detection scan attempts to identify the operating system from network characteristics. For 192.168.200.129, Nmap reports possible Windows-family matches but states that there is no exact OS match because the test conditions are not ideal.

## OS Detection Screenshot




Figure 3. OS detection output from the Nmap scan.

## 6. Identify Open Ports

Record every port shown with STATE = open and the service associated with it.

| Host | Open Port | Service / Detection |
|---|---|---|
| 192.168.200.129 | 3306/tcp | MySQL |
| 192.168.200.2 | 53/tcp | domain / dnsmasq 2.45 |
| 192.168.200.254 | None shown | All 1000 scanned TCP ports filtered |
| 192.168.200.1 | None shown | All 1000 scanned TCP ports closed |

## 7. Security Analysis

| Port / Host | Service | Purpose | Security Risk | Recommended Security Measure |
|---|---|---|---|---|
| **TCP 3306**<br>192.168.200.129 | **MySQL** | Database service used by applications to store and retrieve data. | If unnecessarily exposed, the service may be targeted for unauthorized access or exploitation of weaknesses in the database software or configuration. | Restrict access to trusted systems; use strong authentication; keep MySQL updated; and avoid exposing the database directly to untrusted networks. |
| **TCP 53**<br>192.168.200.2 | **DNS / dnsmasq 2.45** | Provides DNS name-resolution services for systems on the network. | An unnecessarily exposed or poorly configured DNS service can increase the attack surface and may permit unwanted queries or abuse. | Restrict DNS access to intended clients; review the DNS configuration; keep the service updated; and limit unnecessary network exposure. |

The following table presents the security analysis of the open services identified in the provided Nmap results.
**Assessment note:** An open port indicates an exposed service, but the Nmap result alone does not confirm that the service is vulnerable. Configuration, authentication, patch level, and network access controls should be reviewed before making a definitive security finding.

## 8.Screenshots

The document includes the basic scan, service/version scan, and OS detection screenshots in the corresponding sections.

## Ethical Use

Nmap must only be used against systems that you own or have explicit permission to scan. For this task, use a local machine or authorized virtual machine. Do not scan external websites, public IP addresses, production systems, or other people's computers without permission.

## Conclusion

Nmap is a useful network security and auditing tool. This task demonstrates how to identify open ports, services, service versions, and operating-system information on an authorized system and analyze the associated security risks. The scans identified MySQL on TCP port 3306 of 192.168.200.129 and DNS/dnsmasq 2.45 on TCP port 53 of 192.168.200.2, while the other scanned hosts did not show open TCP ports in the reported results. The OS detection for 192.168.200.129 produced several Windows possibilities but no exact match, so the result should be treated as an estimate rather than a definitive OS identification. Overall, the scan provides a useful baseline for reviewing exposed services and applying appropriate access restrictions and updates.